

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

ME 19-205-0403 MECHATRONICS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of sensors and transducers and different mechatronics systems.
CO2: Model and analyse first and second order systems for their time and frequency domain responses.
CO3: Analyse a system for its stability.
CO4: Design mechatronics system for simple applications.
CO5: Use software tools for modeling and simulation of mechatronic systems.

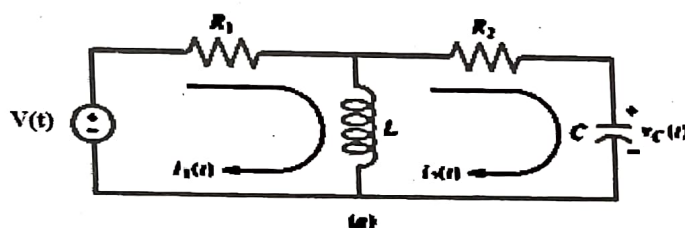
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
L6 – Create
PO – Programme Outcome

(Ordinary graph sheet and semi-log graph sheet need to be provided)

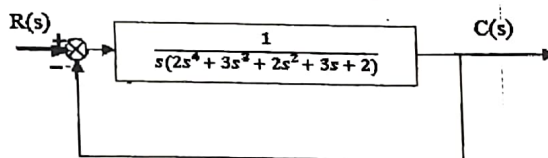
PART A

(Answer ALL questions)

- | | (8 × 3 = 24) | Marks | BL | CO | PO |
|--|--------------|-------|----|----------|----|
| I. (a) Describe the working of an incremental encoder. Explain how the encoder detects direction of rotation. | 3 | L1 | 1 | 1,7,12 | |
| (b) Draw the symbol of a 5 port 2 position direction control valve which is actuated by solenoids on both sides and explain its working. | 3 | L2 | 1 | 1,7,12 | |
| (c) Find the transfer function $G(s) = \frac{I_2(s)}{V(s)}$ of the electrical system shown. | 3 | L3 | 3 | 1,2,6,12 | |



- | | | | | |
|--|---|----|---|----------|
| (d) Find the natural frequency and damping ratio of the second order systems whose transfer function is given by $G(s) = \frac{15}{(s+3)(2s+5)}$. | 3 | L3 | 3 | 1,2,6,12 |
| (e) Check the stability of the unity feedback system shown. Determine the number of poles on the Left Half Plane, Right Half Plane and $j\omega$ axis. | 3 | L3 | 4 | 1,12 |



- | | | | | |
|---|---|----|---|------|
| (f) The transfer function of a system has open loop poles at $s = 0, -2, -4$ and zeros at $s = -1, -3$. Sketch the root locus. | 3 | L2 | 4 | 1,12 |
| (g) Explain the fundamentals of real-time processing in automation and control systems. | 3 | L1 | 4 | 1,12 |
| (h) Explain the working of a micro-processor with a neat sketch of its internal architecture. | 3 | L2 | 4 | 1,12 |

(P.T.O.)

PART B

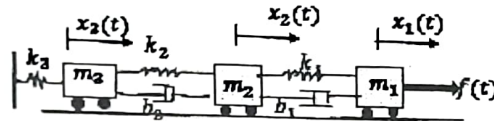
(4 × 12 = 48)

- II. (a) Explain the working of the following
 (i) Rotary Actuator.
 (ii) Stepper Motor.
- (b) A 24 V lamp is to be switched ON when a push button switch is pressed. It has to remain ON even after the push button is released. Design a circuit for this using a 24 V, 2-pole relay.

OR

- III. (a) Explain the principle of a pilot operated valve with the help of neat diagram.
- (b) A single acting pneumatic cylinder positioned vertically is used to lift weights. It is controlled by a three-port two-position direction control valve which has a solenoid on one side and a spring on the other. The cylinder should raise the weight when the solenoid is charged and when the solenoid is off, the weight is lowered due to gravity. Draw the hydraulic circuit diagram for the case.

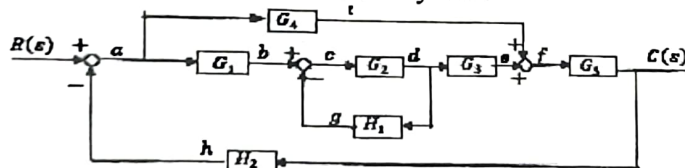
- IV. (a) Obtain the force-voltage systems for the mechanical system shown.



- (b) A first order system has a transfer function $G(s) = \frac{125}{100s + 125}$. What is the time constant of the system? Determine the output of the system after $t = 0.5$ s when it is subjected to a unit step input.

OR

- V. (a) Find the overall transfer function of the system.



- (b) A second order control system has the following specifications.
 (i) Maximum percentage overshoot permitted is 20%.
 (ii) 2% settling time must be less than 2 s.

Find the required values of the natural frequency and damping ratio of the system. Obtain the transfer function. What will be its rise time and peak time?

- VI. Draw the root locus of the system if $G(s)H(s) = \frac{K}{s(s+4)(s^2+2s+2)}$.
 For what value of K, the system becomes unstable?

OR

- VII. Draw the Bode diagram for a unity negative feedback system having a forward transfer function $G(s) = \frac{10(s+2)}{s(s+1)(s+3)}$ and obtain the gain margin and phase margin, if it is possible.

(Continued)

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		Marks	BL	CO	PO
VIII.	(a) Represent the following Boolean algebra equation in a PLC logic ladder diagram $Z = (A.B + C). \bar{D}.E.\bar{F}$.	6	L3	5	1,12
	(b) A washing machine has a pump to pump in water, a heater to heat water and another pump-to-pump water out. Write the ladder program to achieve the following: The pump is switched on to fill water for 3 minutes. Then the water is heated till the required temperature is reached, i.e., till the temperature sensor produce signal. On pressing the switch for the second pump, it pumps out water for 2 minutes and stops.	6	L4	5	1,12
OR					
IX.	(a) A water pump for filling a tank is started using a spring-return push-button switch. When a sensor detects that the tank is full, the pump is stopped. Then the water is heated for 300 seconds using a heater. Draw a ladder diagram to represent this.	6	L4	5	1,12
	(b) The entry and exit of cars into a parking space are restricted by barriers. At the entrance, the driver has to insert a coin in the slot provided. A sensor detects this and the barrier opens (rises). It remains open for 10 seconds for the car to pass, then lowers. At the exit, a switch has to be pressed and the barrier rises. It remains open for 10 seconds and then lowers. There are limit switches at both ends of the entry and exit barriers, which give a signal when the barrier is fully open or fully closed. Draw a ladder logic diagram to achieve the above.	6	L4	5	1,12

Blooms's Taxonomy Level's

L1 - 5%, L2 - 15.8%, L3 - 32.52%, L4 - 46.68%.
